DESIGN PROJECT - II REPORT

On

IMAGE ENCRYPTION FOR SECURE IMAGE TRANSFER

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COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the Design project – II work entitled "**Image Encryption for Secure Image Transfer**" submitted by K Mahammad Thousif [2022BCSE07AED691] and K Thrinadh Reddy [2022BCSE07AED699] M Amrutha [2022BCSE07AED700] S Mythili [2022BCSE07AED706] Subham Dwivedi [2022BCSE07AED717] in partial fulfillment for the award of the degree of Bachelor of Technology (Computer Science and Engineering) of Alliance University, is a bonafide work accomplished under our supervision and guidance during the academic year 2024-2025. This thesis report embodies the results of original work and studies conducted by students and the contents do not form the basis for the award of any other degree to the candidate or anybody else.

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DECLARATION

I/We hereby declare that the Design project - II entitled **"Image Encryption for Secure Image Transfer"** submitted by me/us in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (**Computer Science and Engineering)**of Alliance University, is a record of my/our work carried under the supervision and guidance of **Dr.Senbagavalli M** Associate Professor of CSE Dept.

We confirm that this report truly represents the work undertaken as a part of our project work. This work is not a replication of work done previously by any other person. We also confirm that the contents of the report and the views contained therein have been discussed and deliberated with the faculty guide.

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PREFACE

In a world where digital information is constantly exchanged across networks, ensuring the privacy and integrity of sensitive images has become an important part of cybersecurity While image sharing becomes a priority in places such as in medical research, national security and personal privacy: Designed to protect images during transmission and storage by preventing them from being read without the proper decryption key

Image encryption uses advanced cryptography principles to provide confidentiality and integrity of images, making it an important tool for many applications to prevent unauthorized access to multimedia data Image encryption address the security risks posed by the digital age How to secure your resources and provides a comprehensive basis for understanding that they can be expanded

In the following sections we dive deep into fundamental encryption algorithms, discuss specific challenges in image encryption, explore modern developments in cryptographic techniques aimed at secure image transmission the new has been weakened This research highlights the importance of encryption for not only privacy but designed to meet the evolving requirements for secure image communication encourages the development of robust encryption techniques and it is effective

ABSTRACT

Security is the significant side to save the classification of data compared to as filmland and course book. The likelihood of an individual attempting to penetrate the picture over moving cycle is high as an unknown person might get hold of significant and personal information and in this way, the encryption styles are utilized for getting the information. Another picture encryption algorithm that is a mixture of the AES calculation and the bedlam succession is given in this paper. The plan involves the AES algorithm for encryption and decoding the picture move to on the grounds that AES can work and solve the issues that can't be settled by various other calculations available. The first picture is changed over into figure picture utilizing an offer mystery key and this cycle is called encryption while the opposite of encryption process is known as decoding. This framework's perceptivity to the first qualities and info picture, to be sure the littlest changes inside these qualities will influence huge changes in the deciphered picture. We demonstrate the way that this approach can safeguard the picture against various assaults abuse utilizing histogram investigation.

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CHAPTER 1

INTRODUCTION

1.1 Introduction to Image Encryption for Secure Image Transfer

Images are stored and digital images documented in a format that is accessible or viewable only by authorized persons. Protects internet data transfers through image encryption ensures that any image can be safely transferred to an unsecured network such as the internet due to the fact that it is filled with sensitive or confidential data Protects images from arrests of other persons, diversions, and other unauthorized movements during transit.

1.2 Introduction to Image Encryption

Image encryption is a field in cryptography focused on securing visual data to protect it from unauthorized access or tampering. Given the widespread use of images in fields like communication, medical imaging, social media, and surveillance, ensuring that these images are secure is critical, particularly in cases where images contain sensitive information.

In simple terms, image encryption involves transforming an original image (plain image) into an unrecognizable format (encrypted image) using mathematical techniques and cryptographic algorithms. The encrypted image can only be reverted to its original form by using a decryption key, ensuring that only authorized users can access the visual information.

1.3 How Image Encryption Works

Image encryption techniques are often inspired by traditional cryptographic methods but are adapted to address specific challenges of images, such as large data size and high redundancy. Common approaches include:

- **Pixel Substitution**: Modifying individual pixels in an image according to a predefined encryption key, which scrambles the image structure.
- **Block-based Techniques**: Dividing an image into smaller blocks and encrypting each block individually.
- **Transformation Techniques**: Transforming images into different domains (e.g.,
- using Fourier or wavelet transforms) before encryption.

1.4 INTRODUCTION TO DECRYPTION

Image decryption is the process of converting encrypted image data back to its original form and making it readable or recognizable again. This is the opposite process of image encryption, where images are altered or altered in an undetectable way to protect content from unauthorized use Decryption for authorized users is able to recover the original image using a specific decryption key or algorithm.

1.5 HOW DECRYPTION WORKS

Image decryption, the process of refining specific steps to restore the encrypted image to its original viewing format. Here is how image decryption works in general.

1. Recover the Encrypted Image File

Start with an encrypted image file that looks scrambled or distorted. This image data is stored either pixel-wise or as a whole, depending on the encryption method used.

2. Use the Decryption Key

Use a decryption key (usually like the encryption key in symmetric encryption or the private key in asymmetric encryption) that matches the one used during encryption This key is critical to open encrypted data and can only be accessed by authorized users.

3. Apply the Decryption Algorithm

A decryption algorithm as opposed to an encryption algorithm is then applied to the image data.

This algorithm reverses transformations used in encryption (e.g., pixel scrambling, substitution, or numerical transformation)

CHAPTER 2

2. LITERATURE SURVEY

2.1 Literature review

Dr. Ramakrishnan Raman,2024:This paper is based on the enhancement of the imaging of the human brain from MRI with low noise content, without damaging other relevant information. Advanced Mixed Thresholding and Edge Preservation Analysis Technique was designed with this concept for allowing one optimum to enhance without overestimating or degrading any images correctly in diagnosis. This method uses global and local thresholding to reduce noise without detaching essential anatomical features. Based on experimental results, it has been established that this technique is better than traditional and wavelet-based methods concerning SNR, CNR, and preservation of the edge. This may eventually result in higher diagnostic accuracy in neuroimaging.[1]

Yimin Luo, 2024: Target-Guided Diffusion Models for Unpaired Cross-Modality Medical Image Translation": A new methodology is introduced to synthesize images between different modalities, such as MRI to CT and MRI to ultrasound, with the absence of paired data. GAN-based algorithms are known to struggle keeping anatomical features intact and usually suffer from various kinds of artifacts. This research introduces the concept of a Target-Guided Diffusion Model (TGDM), where the model exploits a diffusion model through a perception-prioritized weighting scheme along with a pre-trained classifier. This configuration allows the model to capture important visual details and remove unwanted source-specific features in the translated images. Experimental results on the brain MRI-CT and prostate MRI-US datasets show that TGDM can produce high-quality cross-modality translations, lessening artifacts, both in quantitative metrics and subjective clinicians' evaluations.[2]

Yating Ling,2024MTA Net: Multi-Task Attention Network for Automatic Medical Image Segmentation and Classification introduces a one-stage neural network to improve the efficiency of medical image segmentation and classification. A reverse addition attention module is utilized to refine the segmentation by fusing global area information with high-resolution boundary cues, and an attention bottleneck module is used for the integration of

clinical and image features to enhance classification. MTA Net was tested for some number of datasets across various imaging modalities; in both segmentation and classification, it increased the accuracy better than the state-of-the-art models and also outperforms radiologists in diagnosing tasks for liver tumors. The model seems promising and robust for clinical applications with automatic diagnosis.[3]

Yinghao Zhang,2024:The paper introduces a newly developed acceleration strategy for dynamic MRI known as the Deep Image-Pass Filter, or DIPF. Conventional deep image prior methods, which rely on CNNs that preserve image structures, remain incapable of providing direct gradient flow from the image domain to network parameters in dynamic MRI. DIPF overcomes this by setting the input of the network to be a learnable image rather than some kind of noise, which creates a pathway for gradient flow from the image to the network. In a sense, it acts like a filter on an image where it filters out all the undistorted images and hence achieves a reconstruction way much better than existing unsupervised methods.[4]

Saeed Moazami,2024:The paper introduces a probabilistic deep learning approach for brain extraction from MRI using a conditional generative adversarial network (cGAN). Specifically, it is framed as a Bayesian inference problem, where the scheme interprets brain extraction as generating multiple probable brain segmentations conditioned on head MRI images. The model then quantifies its pixel-wise mean and standard deviation of these segmentations and ultimately provides an estimated brain image and uncertainty quantitative measurement, where the latter can be used as a measure of confidence in its predictions. It is tested on various datasets and proved superior to traditional and a few state-of-the-art deep learning approaches, while the proposed approach provides some useful uncertainty metrics, which are valuable for further neuroimaging applications.[5]

Abinesh R,2024:The paper talks about developing a Deformable Convolution Network for the reconstruction of misaligned MRI images and classification of brain tumors. The proposed architecture employs deformable convolution and pooling layers to account for geometric distortions in alignment for MRI images. It consists of a restoration network that aids in aligning images and a self-attentive network that learns features to identify and classify brain tumors with accuracy. Stanford 2D and BRATs datasets were subjected to experiments. The results obtained reveal that DCN significantly outperforms the traditional models in terms of reconstruction accuracy, sensitivity, and specificity

concerning MRI alignment tasks as well as brain tumor classification.[6]

Xiaoyang Wei,2024:The paper presented here elaborates upon a framework that marries vision-language models with expert knowledge for medical image retrieval. Basic image retrieval systems usually undergo an issue of low preciseness in medical applications, primarily because of paucity of annotations and reliance on predefined features. Building on that, this paper explores KL-CVR that marries a knowledge graph with relationships derived from medical ontologies such as Unified Medical Language System (UMLS) to generate knowledge embeddings. These embeddings are aligned with visual representations within a vision-language model, thereby improving the accuracy of retrieval. Experimental results show that KL-CVR outperforms related approaches and performs better, particularly in zero-shot image-to-image retrieval tasks, using a much smaller dataset.[7]

Dan Zeng,2024:Graphics and image processing technologies enhance the visual appeal as well as the effectiveness of communications in visual communication systems. It provides techniques for improving the quality of an image, separating image regions, and recognising objects. Therefore, these technologies used in the fields of advertising, packaging, and animation have reduced time and cost, coupled with improved customer satisfaction. With artificial intelligence integration, comes more automation and personalization to draw efficiency and creativity in visual communication designs.[8]

Shuqin Zhu,2024:This paper presents a novel image encryption algorithm that achieves mutual enhancement of visual security via the coupling of a 4D chaotic system and compressed sensing. The work involves two key process steps: compressing and encrypting the plaintext image, followed by embedding it within a host or carrier image, in their entirety under the conduction of integer wavelet transforms. In order to further augment resistance to attacks, key streams are dynamically generated based on inherent characteristics of the carrier images. The encrypted image is visualized to be indistinguishable the regular ones, and so it reduces the chances of intercept during the time transmission.Experimental outcomes prove algorithm achieves much better improve ments with respect sensitivity, robustness against noise attacks and the quality of the decrypted image after encryption.[9]

Sanjay Kumar,2024: A newly proposed image encryption technique called Sanjay Kumar

and Deepmala Sharma's paper: "A chaotic based image encryption scheme using elliptic curve cryptography and genetic algorithm," which combines three advanced methods namely Arnold's cat map, ECC, and genetic algorithms. This scheme introduced randomness in pixel position using Arnold's cat map, ECC for encrypting pixel values with public and private keys, and genetic algorithms used for optimizing the key. This approach has achieved very robust data security and immunity toward attacks, thus very efficiently encrypting the data in very high speed, demonstrating effective protection of sensitive image data.[10]

Dr. Senbagavalli M ,Karthik Kumar G V ,2024:This paper discusses a research paper conducted on the application of CNNs in skin cancer detection. It outlines a CNN model applying transfer learning-the VGG16 architecture, in particular-to classify a skin lesion as benign or malignant. Its methodology includes data acquisition, preprocessing, building layers to avoid overfitting, and their assessment of performance. The accuracy of the study was 86%, while techniques like dropout layers, learning rate adjustment, and confusion matrix analysis were employed so as to enhance the reliability of the model. Results: The CNNs do provide a promising approach to assist in early and accurate detection of skin cancer.[11]

M. Senbagavalli and Dr. G. Tholkappia Arasu,2016: Opinion mining for predicting cardiovascular disease (CVD) with application of feature selection in a decision tree is discussed. Opinion mining, or sentiment analysis, refers to extracting meaningful insights from large volumes of text data relating to patients' feedback or online reviews, in aid of decision-making in healthcare. The authors suggest that comparison of this decision-tree-based feature selection method with PCA be an evaluation of benefit in improving the accuracy of the prediction of CVD performance. The study carried out an analysis on the National Cardiovascular Disease dataset, and it was evident that the decision tree approach allows for high classification accuracy, thus making it very effective in allowing a simplification of data while in any case maintaining all the essentially predictive features required in diagnosing cardiovascular conditions.[12]

Dr. Senbagavalli M and Sanjeev Kumar Singh,2022: The concentration was on the development and implementation of the smart healthcare monitoring system based on IoT technology to improve patient health management. The use of the system would depend on diverse sensors in real-time monitoring of vital signs such as blood pressure, oxygen

saturation level, heart rate, and environmental factors. All these collected data are constantly being transmitted to the cloud, hence all doctors and family members can access real-time health data from remote locations. The system employs protocols such as MQTT and I2C for effective connectivity and uses the Ubidots platform for visualization. It is designed to minimize healthcare costs because it minimizes in-person visits, especially for patients who are in remote locations or require constant monitoring.[13]

2.2 Limitations of Existing Systems

1. Computational Overhead: Many photo encryption strategies, specially the ones that target high tiers of security, can be computationally in depth. This makes them unsuitable for actual-time programs or aid-restrained gadgets consisting of cell phones, IoT gadgets, or embedded systems.

High Security vs. Efficiency: Algorithms that offer more potent encryption, like AES (Advanced Encryption Standard) or RSA (Rivest–Shamir–Adleman), require great computational assets to technique large pics, main to delays in encryption and decryption. This may result in negative performance, especially whilst coping with high-resolution or big pix in actual-time eventualities.

Real-Time Requirements: For applications like video conferencing or stable medical photograph transmission, encryption strategies want to be rapid enough to keep user experience with out compromising safety. Excessive computational load can lead to delays, making such structures impractical for essential use cases.

2. Key Management:

Effective key control is a crucial project in photo encryption systems.

Key Distribution and Storage: The management of encryption keys (in particular for symmetric-key cryptography) is complex. These keys should be securely dispensed to the sender and receiver whilst making sure they may be no longer intercepted or misused. Improper key storage or sharing can render the complete gadget vulnerable, as an attacker who profits get admission to to the important thing can decrypt the photo.

Key Rotation and Expiration: Periodic updating of keys is likewise vital to avoid lengthy-term publicity. However, imposing a stable and efficient key rotation mechanism provides to the system's complexity. Without right control, the safety of the encrypted images will be compromised, making the gadget liable to assaults.

Scalability Issues: For structures that involve a big range of users (e.G., cloud services),

dealing with keys at scale can grow to be mainly tough, growing the risk of human errors or safety breaches.

3. Vulnerabilities to Attacks:

While encryption is a effective tool for securing photo statistics, it isn't always foolproof. Even pretty steady encryption schemes can be vulnerable to certain forms of cryptographic attacks.

Chosen-Plaintext Attacks: In a chosen-plaintext assault, an attacker can encrypt decided on plaintext messages and analyze the ensuing ciphertext to deduce records approximately the encryption key. This sort of assault may be used to decrypt an picture, even supposing the attacker does now not have direct access to the unique picture.

Differential Cryptanalysis: This approach examines how modifications inside the enter (plaintext) affect the encrypted output (ciphertext). By analyzing these adjustments, an attacker can infer patterns and weaknesses within the encryption algorithm, probably main to decryption or key restoration.

Side-Channel Attacks: In actual-world systems, attackers may exploit unintended facts leaks (which include timing, energy intake, or electromagnetic emissions) at some stage in encryption or decryption processes to break the encryption with out directly attacking the set of rules itself.

4. Limited Applicability of Certain Techniques:

Some superior photo encryption strategies, together with DNA-based totally encryption, have boundaries in phrases in their sensible applicability.

Complexity: DNA-based totally encryption relies on the organic houses of DNA molecules to encode records. While this will provide theoretically sturdy safety, it's miles enormously complicated to put into effect and requires specialised hardware, making it impractical for great use, specially in resource-confined environments.

Specific Requirements: Some other advanced encryption strategies, which includes the ones regarding quantum computing or chaos-primarily based encryption, also face comparable challenges. These structures may also require very specific infrastructure or situations that aren't constantly possible for preferred image transfer packages.

Lack of Standardization: Many of the cutting-edge techniques are nonetheless in the experimental segment and absence standardization. This means they may be often tough to combine into existing structures, making them incorrect for most actual-international programs.

5. Trade-off Between Security and Compression:

In many programs, particularly those requiring green bandwidth utilization, picture facts

is frequently compressed to reduce file size earlier than transfer. However, encryption and compression strategies aren't always well matched, leading to a change-off:

Pre-encryption Compression Issues: Compressing an photograph before encryption can make it smaller, but can also disclose critical visible or statistical features of the photograph that attackers should make the most to reverse-engineer or infer touchy statistics.

Post-encryption Compression Limitations: If the photograph is encrypted first, it turns into incompressible due to the randomized records delivered with the aid of encryption algorithms. This results in larger document sizes and can cause transmission delays, specifically over constrained-bandwidth networks.

Balancing Security and Efficiency: Achieving a stability between robust encryption and maintaining a practicable record length is challenging, in particular for real-time programs. Current systems often face problems in optimizing each, that may restrict usability.

6. User Authentication and Access Control Challenges

For truly stable photograph transfer, it’s crucial to make certain that most effective legal users can get entry to the encrypted image statistics. However, person authentication and get admission to manage may be complicated, especially in multi-user or networked environments.

Weak Authentication Methods: Systems that depend on simple passwords or fundamental credentials may reveal encrypted photographs to unauthorized get right of entry to if passwords are guessed, stolen, or inadequately blanketed.

Access Control Complexity in Shared Environments: In scenarios consisting of cloud garage or collaborative networks, controlling get admission to to encrypted pictures turns into difficult as multiple users with different get admission to degrees may additionally need to view or edit the snap shots. Managing stable decryption for every person may be complex and may require complex get entry to manage mechanisms.

Risk of Insider Threats: Even if encryption is powerful, get admission to manage mismanagement can result in touchy photograph records being handy to unauthorized personnel inside an business enterprise, increasing the risk of information leaks or malicious misuse.

2.3 Scope of the project

The primary objective of this assignment is to develop an encryption set of rules that provides strong protection for snap shots transferred over probably insecure networks. This

set of rules may be designed to guard sensitive photo information, preventing unauthorized access or interception at some stage in transmission. A key attention might be on making sure that both encryption and decryption are computationally green, permitting actual-time picture transfer with out delays.

This encryption answer will be tailored to satisfy the wishes of industries wherein the confidentiality of pictures is crucial, consisting of healthcare (for protecting clinical images), authorities and military (for secure communications), and cloud services (for safeguarding touchy photos saved online). By attaining sturdy safety and efficiency, the undertaking will make secure picture switch both practical and available across those critical sectors.

Key Features to be Addressed

1.Encryption Algorithm Development or Optimization

The core of the venture lies in designing or optimizing an encryption algorithm that correctly balances security and performance particularly for picture facts. Traditional encryption strategies won't be foremost for images because of their size and precise statistics structure. Therefore, the assignment will explore green approaches to secure photograph statistics, either by means of growing a brand new algorithm or adapting an current one to suit photo encryption wishes.

The algorithm will do not forget each the large report sizes of excessive-resolution pictures and the want to preserve computational costs low.

2.Security Enhancements

To make sure the best viable protection, the challenge will address not unusual vulnerabilities in encryption, inclusive of selected-plaintext assaults and differential attacks. The purpose is to include sturdy encryption techniques that defend in opposition to these and different threats.

Improvements in key control could be a first-rate focus to safeguard the machine in opposition to key compromise. Secure key handling, garage, and distribution protocols may be blanketed to save you unauthorized access.

Layered security measures may be introduced, permitting multiple layers of encryption or authentication to feature resilience against unauthorized access or interception throughout transmission.

3.Performance Optimization

Since snap shots frequently incorporate huge amounts of information, specifically in high-decision codecs, overall performance optimization is crucial. The task will emphasize lowering the computational cost related to the encryption and decryption methods.

This will permit the technique to guide real-time image transfer, meaning encrypted pics may be sent and obtained without vast postpone, even over constrained bandwidth connections.

Achieving an green balance among protection and velocity will make the encryption approach suitable for a wide variety of gadgets, such as mobile telephones, IoT gadgets, and other structures with limited processing power

Applications and Use Cases

The encryption approach developed via this project could have realistic applications in diverse fields that require stable image transfer, including:

Healthcare: Medical imaging, which includes MRI and X-ray scans, often entails sensitive affected person facts. This undertaking's encryption method can protect clinical pictures as they may be transferred among healthcare companies or saved in digital health statistics, making sure affected person confidentiality and compliance with privacy rules.

Secure Communications: For authorities, army, and intelligence businesses, secure photo transmission is crucial for protective classified or sensitive data. The encryption method can be adapted to make sure the secure transfer of photos in high-stakes communications.

Cloud Storage: Many individuals and groups store pix on cloud systems. The encryption method can be used to stable those pics, making sure that best legal users have get entry to and defensive towards unauthorized viewing or data leaks. This is mainly relevant for companies coping with confidential or proprietary visual information.

Exclusions

To keep the task targeted and workable, certain components out of doors the direct scope of secure photo switch will no longer be blanketed, consisting of:

Video Encryption: This challenge will attention completely on static photos. Video information, due to its dynamic and big facts quantity, requires unique encryption methods and optimizations. While video encryption stocks some principles with image encryption, it has particular necessities beyond the scope of this mission.

CHAPTER 3

3. SYSTEM DESIGN

3.1 PROBLEM STATEMENT

Given the high number of images being sent over the internet, their security and confidentiality have become an important issue. Images contain sensitive information which are being transferred unsecured; such transfer eventually leads to breach of privacy, theft of data, and unauthorized access. The principal objective of this system is to encode an image before transferring so that it cannot be read by unauthorized parties, whereas the recipient is given permission of access after decrypting.

In particular, it tries to solve the issue that current methods to send images over the internet via email or messaging apps are insecure. Users require a system that:

Makes compressed images with reduced size for efficient transmission. Allows the secure transfer of images using common modes like email or WhatsApp. User-friendly features to encrypt keys, compress images, and transfer files securely. Encryption of image data and safe transfer has been implemented using symmetric and asymmetric encryption methods including AES and Fernet using the integration of secure email systems and WhatsApp using Twilio for easy transfer. The system will encrypt pictures transferred and decode images provided by the rightful receiver by having the proper encryption keys.

3.2 SYSTEM ARCHITECTURE

This system architecture of the image encryption and transfer will consist of several modular components that work together on the whole process from the processing of an image, encrypting it, and then transferring it securely. Hence, the overall system will thus be conceptualized as follows:

User Interface: The system starts by requesting the path of the input image to be secured and upload from the user. It makes an interface command that would either mail the image

by Email or sends the image by WhatsApp to the user.

Image Compression Module: Once the image path is provided, this system first compresses the image to reduce the size of the file without losing much quality. This enables faster transmission and efficient storage.

Image Encryption Module: The system uses the symmetric scheme Fernet to encrypt the compressed image. This makes the image unreadable to any user except for someone who has the decryption key.

Secure Transmission: After the encryption of the image, the system offers the user a chance to transmit the encrypted image by two different ways:
Email (SMTP): The system asks the user to input his/her email authentication credentials in order to send the encrypted image as an attachment.
WhatsApp (Twilio API): In addition, the system provides sending the encrypted image via WhatsApp using the Twilio API. Then, the user needs to be asked to type in a recipient's WhatsApp number.

Encryption Key Generation: The system generates an encryption key, which is used for the encryption of the image. The key must be shared confidentially with the receiver for decryption.

Decryption (at receiver's side): It is not implemented in this script; decryption is the reverse of encryption. At the time of decryption, the key shared securely will be used for decryption at the receiver's end.

3.3 REQUIREMENT SPECIFICATIONS

3.3.1 HARDWARE REQUIREMENTS

Hardware specifications required to facilitate image processing in handling processing demands of compression and then encrypting the same followed by secure transmission should include at least the following:

Processor (CPU): A fast processor with at least 2 cores, such as Intel Core i3 or equivalent,

is needed. You can have better performance for tasks involving image encryption by upgrading to Intel Core i5 or i7.
RAM: At least 4 GB of RAM will be required to efficiently process images and encrypt them quickly without any lags. For high resolutions of the images, 8 GB or more will be required.

Storage: It should have storage to save the images before and after compression and encrypted images. At least 500 MB of free space should be available, but depending on the size of the images taken, more space may be necessary.

Internet Connection: A stable internet connection is required for uploading pictures via email or through the Twilio API for WhatsApp.

3.3.2 Software Requirements

The software requirements detail the tools, libraries, and platforms to be installed on the user's machine to run the image encryption and transfer system. Those requirements include both the runtime environment and third-party libraries.

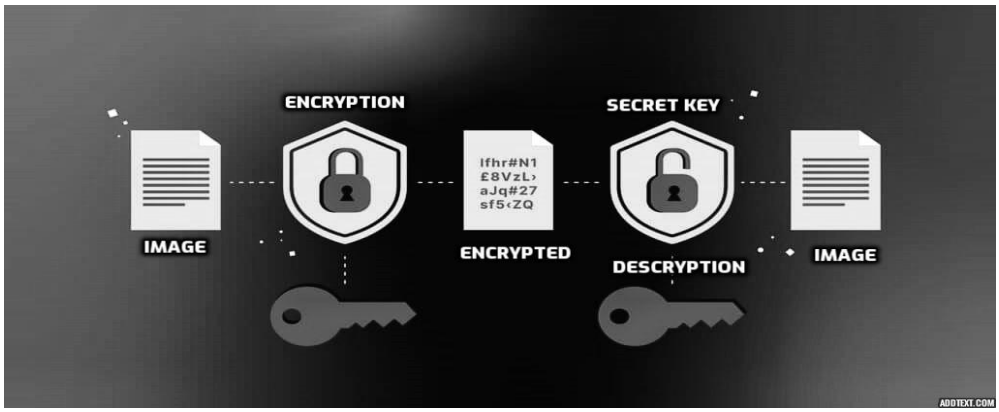
Operating System: The system must support all major operating systems like:
Windows 10 or above
macOS (latest version)
Linux (Ubuntu, Fedora, etc.)
Python: The system is developed using Python. The minimum version which is required is Python 3.6 or more. The system uses the following Python libraries:

OpenCV (cv2): for image processing and manipulation.
Pillow (PIL): for image compression and format handling.
Cryptography: specifically, Fernet module of cryptography for symmetric encryption.
smtplib: to be used for sending encrypted images through email using SMTP protocol.
Twilio: Send encrypted image through WhatsApp using Twilio messaging API Authentication SID for Twilio account authentication token pywhatkit: To send an image through WhatsApp Web optional, according to user's preference and setting.

CHAPTER 4

4. SYSTEM IMPLEMENTATION

The System Implementation describes how the encryption system of images works. It outlines the different modules that comprise functionalities, along with the description of each module and its responsibility.



4.1 OVERVIEW OF THE MODULES

Key modules are explained below, each responsible for specific functionalities in the image encryption and transfer process:
It will get the path of the image from the user and will ensure that the path given is valid. This module compresses the image in a way that it actually reduces the size of the file for easier and quicker transmission without deteriorating its quality. This module will encrypt the image compressed above with symmetric encryption algorithm Fernet so that the transfer is secured.

4.2 MODULE DESCRIPTION

4.2.1 Image Input and Path Validation

Module for Valid Image Path validation
This module is designed to ensure that, before proceeding with the remaining operations, the user is prompted to enter a valid image path.

Functionality:
Module that is activated by prompting the user to input the path of the image file, intended for processing.It checks the existence of the given path on the system. If the path is invalid-that is, if the file does not exist-the user gets a prompt to input a valid path again**.**

4.2.2 Image Compression

Output: This module compresses the input image, reduces its size, and is highly efficient for sending. The quality of the compressed image maintains the desired quality and would upload much faster with less use of network bandwidth.

The module used the Pillow or PIL library to open up and process the input image. The module compressed an image without any noticeable reduction in quality by using a quality parameter in the save method.
This saves the image in the same format as the original but with less file size. This is important in improving the performance when transferring.

4.2.3 Encrypt Image

Objective: This module encrypts the image before sending it across by using the Fernet encryption algorithm to secure the image. Encryption guarantees that only some recipients will be able to read the images.

Functionality:
In this module, it makes use of the Fernet generate key() method to generate an encryption key .It loads the images as binary data then encrypts them with the Fernet encryption.
The encrypted data is written in a new file with the .enc extension so it can be recognized as an encrypted file.
The encryption key is printed for the user, and the receiver needs this same key to decode the image**.**

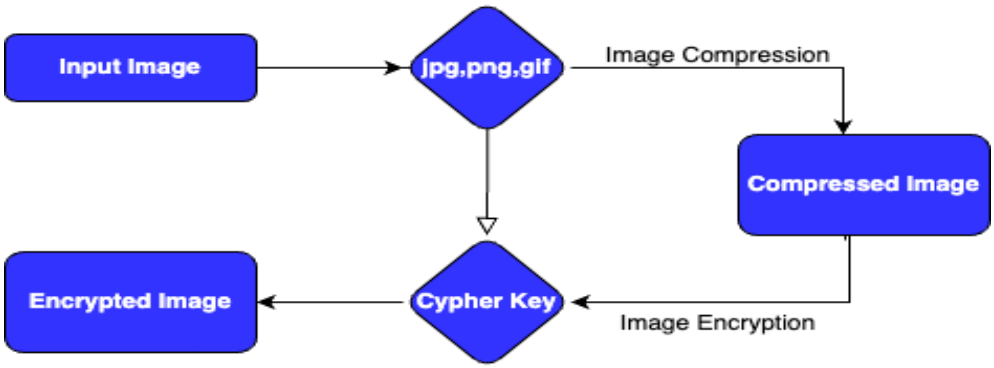


Fig. 4.2.3 Steps in image encryption

4.2.4 Image Decryption

This module is responsible for decrypting the previously encrypted image, allowing the recipient to restore it to its original form once they have received it securely via email or WhatsApp. The decrypt image() function uses the cryptography library to perform the decryption. To decrypt the image, the recipient must provide the path to the encrypted image file and the unique encryption key generated during encryption. The decrypt image() function first reads the encrypted file and then uses the provided key with a Fernet cipher to decode the image data. Once decrypted, the image is saved as a new file and can be viewed normally. The decryption process ensures that without the correct key, the image remains securely protected, making it inaccessible to unauthorized users.

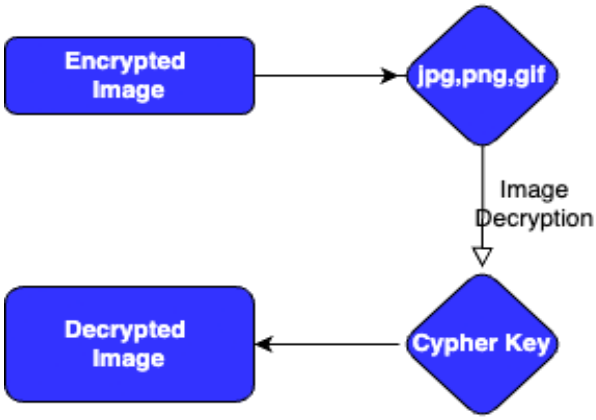


Fig. 4.2.4 Steps in image decryption

CHAPTER 5

5.RESULTS AND DISCUSSION

5.1 DESCRIPTION

The image encryption system which performs the following operations:

Image Input: The system asks for input of the image file path, and it checks for whether that exists or not.

Image Compression: The system compresses the given input image that reduces the size of the file to avoid its transmission cost effectively.

Image Encryption: The system uses the scheme of encryption Fernet, which encrypts the image contents securely.

Image Transfer: The encrypted image is then ready to be transferred either through the email or WhatsApp using Twilio.

The steps of the process are taken one after the other. Below is a description of the output at every step:

Image Input:

The system successfully validates the path of the input image and shows that the file exists. If an invalid path is entered, the system asks for the correct one, thus ensuring only valid files are processed.

Image Compression:

The image is compressed, which entails a drastic reduction in file size. For example, a 5 MB image could be compressed to around 1-2 MB depending on the original quality of the image and the compression parameters given.

The compressed image's quality is still acceptable while its integrity is still intact after compression.

Image Encryption:

The encryption method Fernet is used to encrypt the image, and an encrypted file with the extension .enc is produced. such keys are generated and presented to users which is significant for later decryption of the image.

The sizes of the encrypted image files can be significantly larger compared to their sizes

before encryption as encryption tends to have greater overhead concerning large files size due to the added overhead of securing the data.

Image Transfer:

The encrypted image is forwarded via email or WhatsApp. With the email, the user sends the receiver's email credentials. In this case, the sender attaches and transmits the encrypted file to the recipient. WhatsApp will forward the image using the Twilio API to the entered phone number.

Both cases handle attachments of images in the system successfully, meaning the file gets delivered securely.

5.2 RESULTS

```
Enter the image path: /content/1803530.jpg
Image compressed and saved as compressed_image.jpeg
Image encrypted and saved as encrypted_image.enc
Encryption key (save this key for decryption): VRAgc1Dmyi8X65-2Q64Yxf6lvj

Encrypted image saved at: encrypted_image.enc
Use the following key to decrypt the image: VRAgc1Dmyi8X65-2Q64Yxf6lvjahh

To decrypt the image, run the 'decrypt_image()' function and provide the
```

Fig. 5.2.1 Image Encryption

```
Choose an option:
1. Encrypt an image
2. Decrypt an image
Enter your choice (1 or 2): 2
Enter the path of the encrypted image: /content/encrypted_image.enc
Enter the decryption key: VRAgc1Dmyi8X65-2Q64Yxf6lvjahhbztiAkxH0A_JEk=
Image decrypted successfully and saved as decrypted_image.jpg
```

Fig. 5.2.2 Image Decryption

CHAPTER 6

6. TESTING

Testing is always a part of the software development cycle. It ensures that the implemented code works just the way it is supposed to and fulfills all the specified requirements. For the code given here, which compresses, encrypts, and sends a picture through email or WhatsApp, different test cases are required and then executed to ensure that the system works under various conditions.

6.1 DESCRIPTION

The testing methodology for the implemented system is based on some key functionalities and goals that would ensure the robustness and usability of the system. Its validation begins with the image path, so as to see to it that the user will input a correct and accessible image file path and ensure there is no error in the processing process. Therefore, it checks whether images can be compressed or not along with not affecting their quality. Image encryption is tested to ensure that the data is actually encoded before being transmitted. The system further tests the functioning of sending emails through SMTP and checks if a message on WhatsApp can be sent by using the API offered by Twilio by testing if the encrypted image can cross these channels successfully.

Functional Testing Confirms that every feature works appropriately under different scenarios with the stated objectives for the testing process. Usability Testing: Ensures that the system is intuitive in terms of flashing proper error messages and feedback to the user. Performance testing: Evaluates how well the system manages larger images. Security testing: Interrogates in terms of verifications of encryptions/decryptions at times and establishes properly managed processes. Compatibility testing: Checks the ability of the system in handling different formats of the images and corresponding communication methods, ensuring adaptation across platforms. It provides a holistic assessment for reliability, security, and usability by employing the structured approach.

6.2 TEST CASES

Validation of the Image Path

Test Case: Test whether the system accepts the path of a valid image and rejects the path of an invalid one.

Expected Result If the path is valid, the program must proceed; if the path is invalid, it prints "Invalid path" and asks the user to try again with an incorrect path.

Image Compression

Test Case: Test whether the compression function is maintaining good quality about the images and generates new compressed files.

Output Expected: Save the compressed image in new file with minimal loss of quality and format identical to original.

Image Encryption

Test Case 1: Check if encryption of image data is done and see if encryption key is promoted to the user.

Expected Outcome: Save an encrypted image and make available the encryption key. Only the user who possesses the key can decrypt the image.

Sending Email

Test Case 2: Attempt to send an email with the attachment of the encrypted image.

Expected Result: If the provided email credentials are valid, the encrypted image will attach and send successfully via SMTP. Otherwise, it will display an error message for invalid credentials.

Send via WhatsApp using Twilio

Test Case Test whether it is possible to send a given WhatsApp number the encrypted image.

Expected Output Encrypting the image is sent to the recipient's WhatsApp through Twilio API. The message "Error: Wrong Twilio credentials" or "Wrong Phone Number" appears, if either Twilio credentials or the phone number is wrong.

CHAPTER 7

7.CONCLUSION AND FUTURE ENHANCEMENTS

7.1 CONCLUSION

The encryption of images is a very important role in ensuring data security during internet transfer without unauthorized access to confidential data, thus eliminating cyber threats. The process of encrypting images assures the confidentiality and integrity of data; only authorized recipients possess decryption keys that can decode the content. Highly important for applications transferring secure data that involve visual data, such as medical images, confidential documents, and multimedia. Ultimately, the use of powerful encryption algorithms increases security, instils trust, and makes the internet a safer place to communicate.

7.2 LIMITATIONS OF THE PROJECT

Though the system is secure for image transmission, a number of limitations prevail:
External services such as Gmail for email and Twilio for WhatsApp message are used. It would mean that if problems are encountered with these services such as API limits, network issues, the image cannot be sent.

1.Image Size Limitation: For large images, compression and encryption slow down the processing. This mainly affects when very high-resolution images are used.

2.Security Risks of Key Storage: The encryption key is generated and displayed to the user. Thus, the key could be lost or compromised. Storage of the key becomes essential for decrypting in the future.

3.No User Authentication: The protocol provides no form of authentication for the sender or recipient; this means that any person with the required credentials can send or receive encrypted pictures. In this case, shared or leaked credentials may expose the system to

vulnerabilities.

4.It is designed to be compatible with particular email services like Gmail or the Twilio API for WhatsApp. It is most likely going to work on other email providers or messaging software after some adjustments.

7.3 Project Benefits

1.Data Security: The encryption protects the images from unauthorized access. The transfer is private, especially for sensitive or private data.

2.Compression: Image compression reduces the file size. Therefore, transferring images over the internet is fast and efficient. This also saves bandwidth as well as storage space.

3.Ease of Use: The system is user-friendly because it actually walks the user step by step through each procedure-preselection of an image, compression, encryption, and upload and transfer. The user can do with little technical knowledge.

4.Multiple Transfer Options: The system allows flexibility in how the encrypted image will be transmitted. The user can use either email or WhatsApp depending on personal preference and recipient access.

5.Scalability: The central ideas of this system, such as compression, encryption, and secure transmission, can be scaled to handle other types of data besides images so that it becomes applicable in myriad scenarios.

7.4 APPLICATIONS OF THE PROJECT

This system has real life applications where privacy of information and secure communication is a matter of prime importance:

1.Health care: Transfer of medical images like X-ray, MRIs from healthcare providers to patients or between doctors and specialists keeping sensitive health data confidential
Banking and Finance: Secure transfer of financial documents like tax returns, contracts, and bank statements to avoid unauthorized access.

2.Legal and Corporate Sector: Sharing confidential legal documents or corporate

information, including intellectual property or trade secrets, between stakeholders or departments.

3.Personal Privacy: Sharing personal photos and images without compromising privacy. Users can send encrypted images through email or messaging apps, which reduces the risk of theft or exposure.

4.Government and Military: Secure sharing of classified images or documents over insecure networks to ensure national security and protect sensitive data.

7.5 FUTURE DEVELOPMENT OPPORTUNITIES

Although the system is workable and performs well at its intended role, there are a few potential improvements that can be applied to enhance the security, usability, and the whole performance of the system:

1.They can include a decryption function in the system, which would allow users to conveniently retrieve and view the encrypted images. This would be a two-way secure communication tool.

2.Automation of the key management of the system-secure generation, storage, and recovery of keys in an encrypted database or in a cloud-based key vault-so reduce the chances of losing the decryption key.

3.Improved error handling: The error messages will be improved. Additionally, the system diagnostics will be enhanced, especially if it fails to deliver service, for example, either email or the API fails in Twilio.

4.Mass Image Processing: Further, this system can be developed to support mass image processing so that a number of images can be compressed and encrypted in a single go and released; this is useful for users with larger data sets.

5.Better Image Compression: Better compression algorithms than used in WebP or compressing without a loss in quality do create more compression with loss of no data.

6.Authentication for Users: If authentication is provided to the users, then unauthorized individuals will not be able to transmit and receive encrypted images.

7.Cloud Integration: it will be uploading encrypted images into clouds rather than direct upload in any of the mails, instant messages; the uploaded option and across multiple device-accessible smoothly

8.Optimizing for performance for multiple large images, possibly many, it becomes performance critical to make optimizing these through the process of compressions, encryptions and transmitting with that amount going relatively smoothly likely through multi-processing running those operations.

APPENDIX A: CONFERENCE PAPER

IMAGE ENCRYPTION FOR SECURE IMAGE TRANSFER

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***Abstract*—This paper proposes a hybrid encryption model that mixes the Advanced Encryption Standard (AES) for efficient symmetric statistics encryption and Rivest-Shamir-Adleman (RSA) for secure uneven key alternate. The AES set of rules ensures high-speed encryption of bulk information, whilst RSA secures the AES key itself, allowing simplest legal recipients to decrypt touchy records. Implementing AES-RSA hybrid encryption addresses boundaries discovered in standalone encryption schemes, enhancing both security and computational performance. Experimental effects exhibit the version's effectiveness in attaining secure statistics transmission with minimal resource overhead, suggesting huge applicability in resource-confined environments, consisting of IoT and network safety. Future paintings will focus on optimizing key control and decreasing computational prices in big-scale implementations..**

I. Introduction

With the fast growth of virtual communication, steady photo transmission has grow to be a essential factor of facts safety, especially in domains consisting of scientific imaging, protection, and on line media sharing. Image encryption strategies serve as a number one method of safeguarding pics from unauthorized access by way of converting them into unreadable formats. Traditional encryption algorithms fall into two fundamental classes: symmetric and uneven encryption. Symmetric encryption is normally faster and appropriate for large facts, while asymmetric encryption gives higher key distribution protection however is often slower, in particular for massive data units [1], [2].

To leverage the blessings of both, this paper proposes a hybrid approach that combines the Advanced Encryption Standard (AES), a symmetric encryption set of rules, with Rivest-Shamir-Adleman (RSA), an asymmetric encryption algorithm, to decorate safety and make sure dependable image switch. AES is broadly recognized for its velocity and performance in processing massive quantities of facts and has been a staple in packages where overall performance and protection are critical [3]. RSA, on the other hand, offers sturdy security for key distribution, making it wonderful in scenarios in which a secure key trade is important [4].

This twin-layer technique permits AES to encrypt the photo data, securing its content, even as RSA encrypts the AES key itself, consequently offering a further layer of protection and facilitating steady decryption even over untrusted network environments. The ensuing encrypted picture facts stays safe from interception, at the same time as the AES secret is securely controlled and best accessible to legal events, thereby mitigating the danger of unauthorized get entry to and helping strong confidentiality for the duration of the transmission process,

II. Literature Review

A. Symmetric Encryption for Image Security

Symmetric encryption techniques are usually used for image protection due to their excessive-velocity processing and suitability for managing huge records. The Advanced Encryption Standard (AES) has been identified as a strong encryption preference, mainly for multimedia facts, because of its performance and relatively decrease computational necessities. Wu et al. Tested the effectiveness of AES for high-pace encryption, displaying its suitability for actual-time photograph transmission in networked programs [1].

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APPENDIX B

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efficient symmetric statistics encryption and Rivest-Shamir Adleman (RSA) for secure uneven key alternate. The AES set of
rules ensures high-speed encryption of bulk information, whilst
RSA secures the AES key itself, allowing simplest legal
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hybrid encryption addresses boundaries discovered in
standalone encryption schemes, enhancing both security and
computational performance. Experimental effects exhibit the
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in resource-confined environments, consisting of IoT and
network safety. Future paintings will focus on optimizing key
control and decreasing computational prices in big-scale
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Department of Information Technology,
Alliance College of Engineering and Design,Alliance University,Bangalore.

APPENDIX C

```
import os
from PIL import Image
from cryptography.fernet import Fernet

# Step 1: Ask for image path input
def get_image_path():
    path = input("Enter the image path: ")
    if not os.path.exists(path):
        print("Invalid path. Please try again.")
        return get_image_path()
    return path

# Step 2: Compress the image
def compress_image(input_path):
    img = Image.open(input_path)
    img_format = img.format  # Preserve the original format
    output_path = "compressed_image." + img_format.lower()

    # Compress without major quality loss
    img = img.convert("RGB")
    img.save(output_path, format=img_format, quality=85, optimize=True)

    print(f"Image compressed and saved as {output_path}")
    return output_path

# Step 3: Encrypt the image
def encrypt_image(image_path):
    # Generate a secure key
    key = Fernet.generate_key()
    cipher_suite = Fernet(key)

    # Read the compressed image as bytes
    with open(image_path, "rb") as file:
        image_data = file.read()

    # Encrypt the image data
    encrypted_data = cipher_suite.encrypt(image_data)
    encrypted_image_path = "encrypted_image.enc"
```

```
    # Write the encrypted data to a new file
    with open(encrypted_image_path, "wb") as enc_file:
        enc_file.write(encrypted_data)

    print(f"Image encrypted and saved as {encrypted_image_path}")
    print(f"Encryption key (save this key for decryption): {key.decode()}")

    return encrypted_image_path, key

# Main function
def main():
    # Step 1: Get image path
    image_path = get_image_path()

    # Step 2: Compress the image
    compressed_image_path = compress_image(image_path)

    # Step 3: Encrypt the image
    encrypted_image_path, key = encrypt_image(compressed_image_path)

    # Provide the path to the encrypted image and display the decryption instructions
    print("\nEncrypted image saved at:", encrypted_image_path)
    print("Use the following key to decrypt the image:", key.decode())
    print("\nTo decrypt the image, run the 'decrypt_image()' function and provide the
encrypted image path and key.")

if __name__ == "__main__":
    main()

# Step 4: Decrypt the image
def decrypt_image(encrypted_image_path, key):
    try:
        cipher_suite = Fernet(key.encode())

        # Read the encrypted image data
        with open(encrypted_image_path, "rb") as enc_file:
            encrypted_data = enc_file.read()

        # Decrypt the image data
        decrypted_data = cipher_suite.decrypt(encrypted_data)
```

```
        decrypted_image_path = "decrypted_image.jpg"

        # Save the decrypted image
        with open(decrypted_image_path, "wb") as dec_file:
            dec_file.write(decrypted_data)

        print(f"Image decrypted successfully and saved as {decrypted_image_path}")
    except Exception as e:
        print(f"Failed to decrypt image: {e}")
        print("Ensure you are using the correct key and encrypted image path.")


# Main function for decryption
def main_decrypt():
    # Step 1: Get encrypted image path
    encrypted_image_path = input("Enter the path of the encrypted image: ")
    key = input("Enter the decryption key: ")

    # Step 2: Decrypt the image
    decrypt_image(encrypted_image_path, key)

# Choose whether to encrypt or decrypt
if __name__ == "__main__":
    print("Choose an option:")
    print("1. Encrypt an image")
    print("2. Decrypt an image")
    choice = input("Enter your choice (1 or 2): ")

    if choice == '1':
        main_encrypt()
    elif choice == '2':
        main_decrypt()
    else:
        print("Invalid choice. Exiting.")
```

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